INCH-POUND

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MILITARY SPECIFICATION

CLOTH, INSULATION, MULTIPLE LAYER COMPOSITE, QUILTED, FLAME RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers a quilted composite insulation cloth made up of layers of coated glass fabric, flame resistant polyester batting, and metallized film.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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SPECIFICATIONS

FEDERAL

V-T-285 - Thread, Polyester
PPP-P-1136 - Packaging of Coated (Plastic; Rubber) and Laminated Fabrics

STANDARDS

FEDERAL

FED-STD-751 - Stitches, Seams, and Stitchings
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1487 - Glossary of Cloth Coating Imperfections

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies are available from the Federal Trade Commission, Pennsylvania Avenue at Sixth Street, N.W., Washington, DC 20580-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C 518 - Steady-State Heat Flux Measurement and Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus
D 578 - Glass Fiber Yarns
D 579 - Greige Woven Glass Fabrics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material and components It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2 1 Cover fabric.

3.2.1.1 Base cloth. The base cloth shall meet the requirements of ASTM D 579, except it shall consist of either ECC 150 1/0 or ECD 225 1/0 fiberglass yarns as designated in ASTM D 578, in plain weave construction with a maximum weight of 4 0 ounces per square yard (see 4.4.2).

3.2 1 2 Coating compound. The coating compound shall be an approved acrylic or urethane-acrylic aqueous polymeric system which is coreactive with water repellent silicons and flame retardant elements (see 6.5). The compound shall be compatible with the base cloth to ensure its proper adherence to the yarns of the base cloth.

3.2 1.3 Coated cover fabric. The coating compound shall be applied to either or both sides of the base cloth in such a manner so that the coated cover fabric shall conform to the requirements of table I when tested as specified in 4.4.3.

TABLE I. Physical requirements of coated cover fabrics

Characteristics	Requirements
Weight, finished, oz/yd^2(max)	6.0
Width, finished, in	54.0 $\pm$ 1.0
Yarns per inch (min):	
Warp	52
Filling	46

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TABLE I. Physical requirements of coated cover fabrics (cont'd)

Characteristics	Requirements
Breaking strength, lbs. (min):	
Warp	110
Filling	110
Creased breaking strength, lbs. (min):	
Warp	40
Filling	40
Low temperature resistance	1/
Spray rating scale rating (min)	70
Hydrostatic resistance, cm (min):	
Initial	75
After low temperature exposure	45
Flame resistance:	
After flame, seconds (max)	2
Percent consumed, percent (max)	30
Flexibility, in-lb (max):	
Initial	0.010
Low temperature	0.040
Topstitched seam efficiency, pounds (min)	88
Abrasion resistance,	Pass 1/ 2/

1/ The specimen shall not crack.

2/ There shall be no evidence of breakage or cracking on any specimen when tested as specified in 4 4.3

3.2.1.4 Color. The color of the finished cloth shall be natural or white

3.2.1 5 Face identification. The face of the coated cloth shall be stamped "Face" with indelible ink at the beginning and end of the roll.

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3.2.2 Flame resistant polyester batting.

3.2.2.1 Fiber The fiber used in fabricating the batting shall be made of inherently flame resistant virgin polyester manufactured from polyethylene teraphthalate resin modified by phosphorous compounds which are reacted and incorporated into the polyethylene teraphthalate polymer chain. These fibers shall not support combustion. The batting fiber blend shall be 75 percent of 6 denier, 2-1/4 inches ± 1/4 inch staple length fibers with 10 crimps ± 1 crimp per inch, and 25 percent of 15 denier, 3 inches ± 1/8 inch staple length fibers with 9 crimps ± 1 crimp per inch. The fibers shall have a minimum melting point of 478°F with a water soluble antistatic finish not exceeding 1 percent of fiber weight (see 4.4.2).

3.2.2.2 Bonding agent and application. The flame resistant polyester fibers specified in 3.2.2.1 shall be bonded with durable a flame retardant treated bonding agent uniformly applied to both sides. The concentration of the applied bonding agent shall be 18 ± 2 percent on the weight of the finished bonded cut staple batting (see 4.4 2).

3 2.2.3 Finished bonded batting. The bonded fibers in the finished cut staple flame resistant batting shall be well-opened, uniform, and separated without excessive breakage. The batting shall show no neps or multiple fiber ends The finished batting shall conform to the physical properties listed in table II when tested as specified in 4.4.3.

TABLE II. Physical properties of flame resistant polyester batting

Characteristaics	Requirement
Weight, oz/yd^2	2.0 ± 0.2
Width, in.	54.0 ± 1.0
Thickness, in. @ 0.01 psi (min)	0.30
Batting evenness, oz/yd^2	± 0.3 1/

1/ The evenness of the batting shall be such that no single weight determination deviates more than 0.3 oz/yd^2 from the mean of five specimens tested as specified in 4.4.3.

3.2.3 Metallized reflective film. The metallized reflective film base shall be a low density polyethylene film base with 99 9 percent pure aluminum vapor deposited on the base having a surface resistivity of 2 to 3 ohms per square prior to top coating (see 4.4.2). The metallized film shall be topcoated with an electron beam, curable, colorless, acrylic epoxy or an equal scuff resistant coating to provide abrasion resistance for the aluminum deposition. The film shall conform to the physical properties listed in table III when tested as specified in 4.4.3

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TABLE III Physical requirements of finished metallized reflective film

Characteristic	Requirement
Weight, oz/yd^2 (min)	0.4
Width, finished, in.	54.0 ± 1.0
Thickness, mils	0.75 to 1.25 ± 0.05

3.2.4 Thread The thread used for the needle and bobbin in the quilting fabrication of the multiple layer composite shall conform to type I, class 1, subclass B, size E/3 ply of V-T-285

3.2 4.1 Thread color. Two thread colors shall be used for identification of the two panels which make up the total composite. The thread color for the quilting of the outer panel (see figure 3) shall be Olive Drab S-1, C A. 66022. The thread color for the quilting of the inner panel (see figure 3) shall be natural.

3 3 Quilted composite

3 3.1 Panels. The inner and outer panels shall be separately fabricated as shown in figure 1 using the zigzag pattern of quilting stitching shown in figure 2 such as to conform to figure 3.

3.3.2 Cloth alignment. The coated cover fabric specified in 3.2 1 3 shall be orientated in the quilted panels so the identified fabric face is on the exterior facing outward. The cloth covering, layers of polyester batting and metallic film for each panel shall be positioned as to yield a straight edge on the side of the ensemble. The batting and film components shall be flush or extend beyond the cloth covering selvages on both sides of the ensemble. Any extension of the materials on the straight edge shall not exceed 1/4 inch wide while the extension of materials on the opposite side shall not exceed 1 inch The extending edges of the filler materials shall be evenly trimmed without ragged edges.

3.3.3 Stitching and stitch pattern. All stitching shall conform to type 301 or type 401 of FED-STD-751. The quilt pattern shall be a zigzag pattern conforming to figure 2 with 4 to 6 stitches per linear inch in the 6-inch pattern repeat. The stitch pattern for each panel shall be evenly offset so that stitching rows do not overlap when the two panels are placed batting sides together, and used as one composite (see figure 3). The distance between zigzag rows shall be 10 inches ± 1/4 inch. All panel component layers shall be sewn together at the selvage area to facilitate handling the multiple component layers in the end item edge binding and seaming operations. Two color threads shall be used in stitching for identification purposes. The stitching on the outer panel shall be Olive Drab S-1, while the quilt stitch on the inner panel shall be natural

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3.3.3.1 Thread breaks and open stitching. Thread breaks and open stitches shall be secured by stitching back of the break in conformance with the pattern, not less than 1 inch.

3.3.4 The composite configuration. The inner and outer panels of this composite shall be placed batting sides together and must be utilized as one composite. The composite of outer cloths, batting layers, and metallic film shall conform to the requirements of table IV, when tested as specified in 4.4.5.

TABLE IV. Physical and thermal properties of the finished quilted composite

Characteristic	Requirement
Composite weight: oz/yd^2 (max)	23.0
Width, finished: in	54.0 ± 1.0
Composite thickness: in @ 0.002 psi (max)	2.0
Composite R value $R=^{o}F\ Ft^2 \cdot Hr \cdot Btu^{-1}$	3.2
Flame resistance, after flame: sec. (max)	6.0

3.4 Length and put-up. All quilted composites shall be furnished without fabric splices (joining of fabric outer coverings) and without laps or gaps in the batting and film fillers. Unless otherwise specified (see 6.2), quilted multiple layer composites, purchased directly by the Government, shall be furnished in continuous lengths of 61 yards ± 1 yard. Each length shall be put-up on a roll as specified in 5.1.

3.5 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.6 Workmanship. The quilted multiple layer composite shall conform to the quality of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b Quality conformance inspection (see 4.4).

4 3 First article inspection. When a first article is required (see 3 1 and 6.2), it shall be examined for the defects specified in 4.4 4 1, 4 4.4 2, and 4 4.4.3 and tested as specified in 4.4.5.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4 4.2 Certification. Components listed in table V shall be inspected on the basis of a contractor's certificate of compliance with the indicated requirements.

TABLE V. Component certification

Component	Characteristic	Requirement paragraph
Base cloth	Yarn, weight, weave, yarn count	3.2.1 1
Flame resistant polyester fiber	Material identification 1/, crimp, denier, length of staple, melting point, anti-static finish	3.2.2.1

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TABLE V. Component certification

Component	Characteristic	Requirement paragraph
Bonding agent	Resin trade name Percent on bonded batting	3.2.2.2
Metallic film	Percentage pure aluminum Surface resistivity	3.2.3

1/ The certification shall include the fiber manufacturer's identification data.

4.4.3 Component testing. In addition to the quality assurance provisions of referenced documents, testing shall be performed on components and materials listed in table VI for the characteristics shown. Test methods cited shall be in accordance with FED-STD-191, except where otherwise specified. The sample unit for the cover fabric, the polyester batting, and the reflective film shall be 3 continuous yards each, full width. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE VI. Component tests

Component	Characteristic	Requirement paragraph	Test method
Coated cover fabric	Weight	3.2.1.3	5041
	Width	3.2.1.3	5020 1/
	Yarns per inch	3.2.1.3	5050
	Breaking strength	3.2.1.3	5100 2/
	Creased breaking strength	3.2.1 3	4.5.3
	Low temperature resistance	3.2.1.3	4.5 4
	Spray rating	3.2.1.3	5526
	Hydrostatic resistance:		
	Initial	3.2.1.3	5514
	After low temperature exposure	3.2.1.3	4.5.5
	Flame resistance:		
	After flame	3.2 1.3	5905
	Percent consumed	3.2.1.3	5905

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TABLE VI Component tests

Component	Characteristic	Requirement paragraph	Test method
Coated cover fabric (cont'd)	Flexibility: Initial and at low temperature	3.2.1.3	5202 3/
	Topstitch seam efficiency	3.2.1.3	5110 4/
	Abrasion resistance	3.2.1.3	5302 5/
Flame resistance batting	Weight	3.2.2.3	5041
	Width	3.2.2.3	5020 1/
	Thickness	3.2.2.3	5030 6/
	Evenness	3.2 2.3	5041 7/
Metallized film	Weight	3.2.3	5041
	Width	3.2 3	5020 1/
	Thickness	3 2 3	5030

1/ Method 5020, short specimen method

2/ Except that 10 specimens shall be tested in each of the warp and filling directions and a CRE apparatus with rubber faced jaws be used with a pulling speed of 0.5 inch per minute.

3/ Except that five specimens shall be tested in each of the warp and filling directions The specimens shall be evaluated initially at standard conditions. New specimens shall be prepared for low temperature flexibility and both the specimens and the instrument sahll be conditioned at minus $20^{\circ} \pm 5^{\circ}$F for at least 1 hour and tested at that temperature as specified in Method 5202.

4/ Method 5110, except that each test specimen shall be a single layer of fabric prepared by topstitching with stitch type 301, 8 inches from the specimen edge parallel to the warp yarns. Topstitching shall be 7 to 9 stitches per inch with thread conforming to V-T-285, type I, class 1, subclass B, size E/3 ply and shall utilize a needle measuring a minimum of 0.044 ± 0.001 inch across the blade at the eye. For the purposes of this test, the topstitching shall be considered a seam

5/ Method 5302 utilizing grit 0 emery paper abradant, air pressure under the diaphragm controlled at 3 pounds per square inch (psi), 2-1/2 pound vertical pressure, and a multidirectional rubbing action. The fabric specimen shall be subjected to 350 abrasion cycles, then visually examined for the affect of abrasion. The fabric will be rated "failure" if there is evidence of breakage or cracking after the 350 abrasion cycles.

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6/ Except that the specimens shall be measured at a load of 0.002 psi

7/ Except that the results shall be reported for each of the five specimens as the deviation from the mean of the weights of the five specimens. The size of each of the five weight specimens shall be 12 by 12 inches square

4.4.4 End item examination.

4.4.4.1 Yard-by-yard examination. The cloth shall be examined on both sides for the defects listed in table VII. The same yardages shall be given a through-light inspection for pinholes and lightly coated areas. The through-light inspection shall be performed in accordance with MIL-STD-1487. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one for each warpwise yard or fraction thereof in which it occurs. The lot size shall be expressed in units of yards. The sample unit shall be 1 linear yard. The inspection level shall be II (see 6.5) The number of rolls from which the sample yardage is to be selected shall be in accordance with table VIII. The sample yardage shall be apportioned equally among the selected rolls.

TABLE VII. Yard-by-yard examination defects

Examine	Defect	Classification Major	Minor
Cover fabric	Material not as specified	101	
	Color not as specified, uneven or mottled	102	
	Any smash	103	
	Any cut, hole, tear, scratch, or abrasion mark	104	
	Spot or stain, clearly visible 1/		201
	Abrasion resulting in a weak place or any weak place	105	
	Floats and skips:		
	- Multiple, 3/4 inch or more in combined warp and filling directions	106	
	- Multiple, less than 3/4 inch in length in combined warp and filling directions or single float or skip extending over more than one warpwise or fillingwise inch		202

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TABLE VII Yard-by-yard examination defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Cover fabric (cont'd)	Broken or missing yarns (warp and filling):		
	- Three or more contiguous, regardless of length	107	
	- Two or more contiguous missing for 1 inch or more	108	
	- Two or more contiguous missing for less than 1 inch		203
	Single yarn missing for 4 inches or more	109	
	Crease or wrinkle, embedded	110	
	Five or more kinks, knots, or loops in 9 linear inches, clearly visible 1/ and protruding from the surface of the fabric		204
	Any tight warp section resulting in waviness or dimensional distortion of the fabric	111	
	Any cut, broken, torn, folded or rolled selvage	112	
	Any stringy or loopy, scalloped, tight, or slack selvage		205
	Any pit, blister, tunnel or delamination of coating	113	
	Any lump or heavily coated area		206
	Any embedded foreign matter		207
	Any scorch or burn	114	
	Any tackiness (coating will adhere and not unroll readily)	115	
Thread, polyester	Not color specified or not within established tolerances		208
Workmanship	Stitching pattern not within established tolerances by 1/4 inch or less		209
	Stitching pattern not within established tolerances by more than 1/4 inch	116	
	Thread ends not trimmed		210

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TABLE VII Yard-by-yard examination defects (cont'd)

Examine	Defect	Classification Major	Minor
Workmanship (cont'd)	One or more rows of stitching omitted	117	
	Needle chews	118	
	Loose or irregular tension		211
	Stitch type not as specified	119	
	Broken or missing stitches 1 inch or more (not repaired)	120	
	Broken or missing stitches less than 1 inch (not repaired)		212
	Average stitches per linear inch on 6-inch repeat, less than four or more than six	121	
	Repaired thread breaks or open stitches stitched back less than 1 inch		213
	Face side of coated fiberglass cover fabric not orientated to exterior	122	

1/ At normal inspection distance of approximately 3 feet when viewed against a black background.

4.4.4.2 Overall examination. The cloth shall be examined for the defects listed below Each defect listed shall be counted not more than once in each roll examined. The sample size shall be the applicable number of rolls indicated in table VIII. Each roll in the sample shall be examined over its entire length (see 6.5).

Defects

Overall uncleanness
Objectionable odor
Edges of bonded batting not evenly trimmed
One edge not straight
Batting recessed or extends more than 1/4 inch on straight edge or batting on opposite side extends more than 1 inch from cloth selvages
Constructed batting exhibits overall uneveness or contains cloth distortions or bulges
Width not as specified
Lapping or otherwise joining of ends of batting or splicing (joining of ends of cover cloths) to make a continuous length
Gaps or bare area in polyester batting material

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TABLE VIII. Sample size

Lot size (yards)	Sample size (rolls)
Up to 1,300 1/	3
1,301 up to and including 3,200	5
3,201 up to and including 8,000	7
8,001 up to and including 22,000	10
22,001 up to and including 110,000	15
110,001 and over	25

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined

4.4.4 3 Length examination. The sample for length examination shall consist of those rolls selected for overall examination in accordance with 4.4.4.2. Each roll in the sample shall be examined for continuous length. Any continuous length found to be less than the specified minimum or more than the specified maximum shall be scored as a defect The lot shall be rejected if the total of actual lengths of the rolls in the sample is less than the total of the lengths marked on the tickets (see 6.5).

4 4.5 End item testing. The end item shall be tested for the characteristics indicated in table IX. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table IX shall be followed. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 5 yards of the finished quilted composite The lot shall be rejected if one or more sample units fail to meet any test requirement specified. The lot size shall be expressed in units of 1 yard. The sample size shall be in accordance with the following.

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE IX End item tests

Characteristic	Requirement paragraph	Test method	No. of determinations per sample unit	Results reported to nearest
Weight	3.4.4	5041	5	0.1 oz/yd^2
Thickness	3.4.4	5030 1/	5	0.1 in
R Value	3.4.4	4.5.2	2	0.1 $^oF \cdot Ft^2 \cdot Hr \cdot Btu^{-1}$
Flame resistance, after flame	3.4.4	4.5.1	5	1 second

1/ Except the load shall be 0.002 psi.

4.4.6 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1136.

4.5 Methods of inspection.

4.5 1 Flame resistance test.

4.5.1.1 Preparation of specimens. Cut a sample from each outer and inner quilted panel with dimensions of 54 inches wide by 12-1/2 inches long, such that the outer panel quilt line is approximately 10 inches in from the left selvage and the inner panel quilt line is 5 inches in from the left selvage. Place panels batting sides together, line up left selvage and stitch along top and sides 1/4 inch in from the edge. Five specimens shall be marked on the outer panel. To mark each specimen, draw parallel lines in the warp direction 2-3/4 inches apart with the quilted zigzag stitching positioned between these lines. Repeat these markings at each row of quilting until 5 specimens are marked. Stitch along each parallel line and then across the entire bottom edge of the sample. To cut specimens out of the sample, cut 1/8 inch away fom parallel lines along the entire length to yield 5 test specimens each approximately 2-3/4 inches wide by 12 inches long.

4.5.1.2 Procedure. Evaluate flammability in accordance with Method 5905 of FED-STD-191. The specimens shall be evaluated in the warp direction only.

4.5.1.3 Results. Average the after flame results of the five specimens and report.

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4.5 2 R value test.

4.5.2.1 Preparation of specimen. The dimensions of the test specimens shall be dictated by the heat flow meter. When evaluating the quilted composite, the total composite shall be made up of the inner and outer panel and shall be evaluated with batting sides together. The test specimen shall be sampled such that a quilted portion of the composite shall be included in the metering area of the heat flow meter.

4.5.2.2 Procedure. The test specimen shall be evaluated in accordance with ASTM C 518 with the hot plate (T_h) at 120°F and the cold plate (T_c) at 72°F. The hot and cold plates shall contact the specimen at a static load of 0.002 psi.

4.5.2.3 Results. The results shall be reported in thermal resistance units of $^{o}F\ Ft^{2}\ Hr\ Btu^{-1}$ which is expressed as "R" value.

4 5 3 Breaking strength after creasing test Twenty 8 by 4-inch specimens shall be cut from each sample unit. Ten specimens shall be cut with the long dimension in the warp direction and ten with the long dimension in the filling direction No two specimens shall contain the same warp yarns for the warp test nor the same filling yarns for the filling test. No specimens shall be taken nearer the selvages than 1/10 of the fabric width Loop each specimen end to end with the back sides of the fabric together. Do not flatten the loop by hand. Each specimen shall be creased by rolling a steel cylinderical 5-pound weight parallel to and centered over the loop. Push the weight so that it rolls smoothly and straight over the looped area without applying any pressure other than the weight of the roll. Roll the weight at a rate in which it will transverse the specimen in approximately 1 second A single roll across the specimen will be one pass and shall be repeated until five passes have been applied The creased specimen shall be tested for breaking strength in accordance with Method 5100 of FED-STD-191, except the specimen shall be 8 by 4 inches and the pulling speed will be 0.5 inch per minute using a CRE apparatus with rubber faced jaws. Unfold the creased specimen, being sure not to fold backwards beyond 180 degrees. Position the creased area midway between the clamps of the tensile testing apparatus and test until the specimen breaks. The creased breaking strength shall be the average of the results obtained from the specimens tested in the warp and filling directions and shall be reported separately.

4.5.4 Low temperature resistance test. Method 5874 of FED-STD-191 shall be used except that five specimens shall be tested from each sample unit, the specimens shall be exposed to - 20° ± 5°F for not less than 4 hours, and the weight of the steel roller shall be $\overline{5}$ pounds.

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4.5.5 Hydrostatic resistance after low temperature exposure test. This test shall be conducted only for lots meeting the low temperature resistance requirements in table I. The same specimens tested in accordance with 4.5.4 shall be tested for hydrostatic resistance in accordance with Method 5514 of FED-STD-191. The water shall be applied to the face side and creased area of the specimen. The height of the column at which any specimen fails shall be reported to the nearest 0.5 cm. The lot shall be rejected if any specimen fails to meet the specified requirement

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Levels A and Commercial. The multiple layer cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1136.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5 2.1 Levels A, B and Commercial. The multiple layer cloth shall be packed in accordance with the applicable requirements of PPP-P-1136

5 3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1136.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6 1 Intended use. The quilted composite, consisting of the inner and outer panels assembled batting sides together, is intended for use in the manufacture of insulated tent liners

6 2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Length, if other than specified (see 3.4)
e. Levels of put-up, preservation, and packing (see 5.1 and 5.2).
f. Acceptance criteria required (see 6.5).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Approval of coatings. Approval of coatings used on the cover fabric specified by this document is the responsibility of the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 07160-5014. Approval is based on extensive evaluations, including those for toxicity, which are not set forth in the document. Because of the time necessary to conduct a full evaluation, only those bonding agents and coatings already approved and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement.

6.5 Acceptance criteria. The acceptance criteria below are recommended for use. The acceptance criteria as specified in the contract or purchase order shall be binding. Unless otherwise specified, the following acceptance criteria are in accordance with MIL-STD-105

6.5.1 For yard-by-yard examination. An acceptable quality level (AQL), expressed in terms of defects per hundred units, of 2.5 for major defects and 6.5 for total (major and minor combined) defects is recommended

6.5.2 For overall and length examination. The following acceptance numbers are recommended.

For sample size (rolls)	Acceptance number
Up top 15	0
25	1

6 6 Subject term (key word) listing.

Batting
Cold weather
Tent liners

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

Army - GL
Navy - NU
Air Force - 99

Review activities:

Army - MD
Air Force - 82
DLA - CT

Preparing activity:

Army - GL

(Project 8305-0289)

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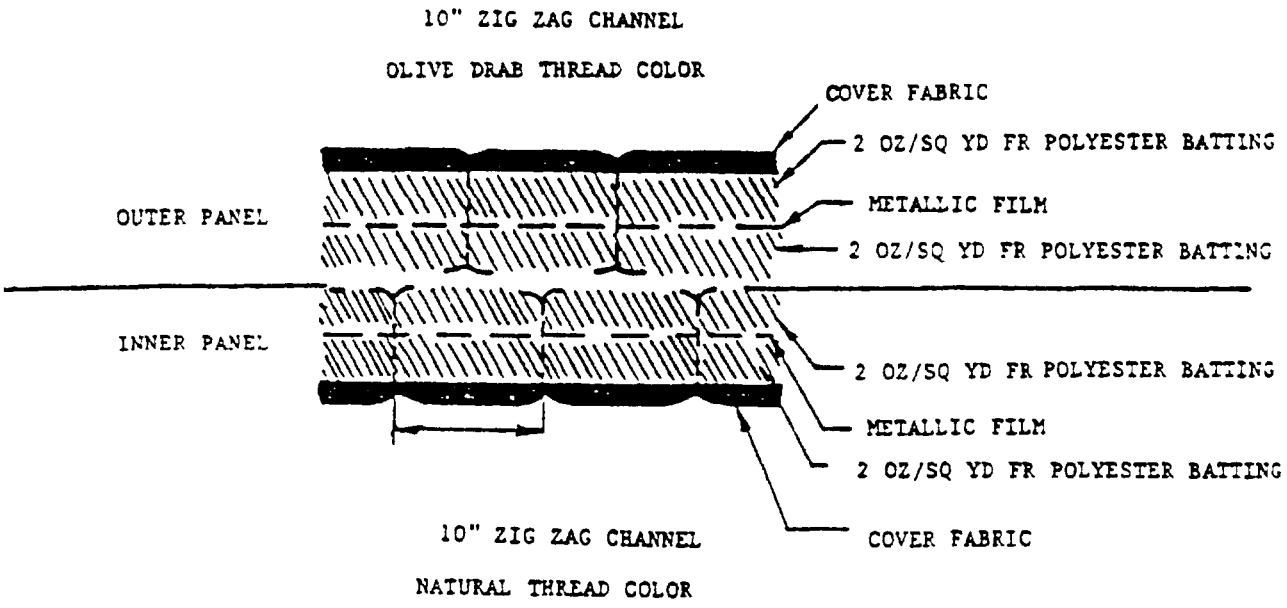


FIGURE 1. Schematic cross section of cloth, insulation, multiple layer composite, quilted, flame resistant.

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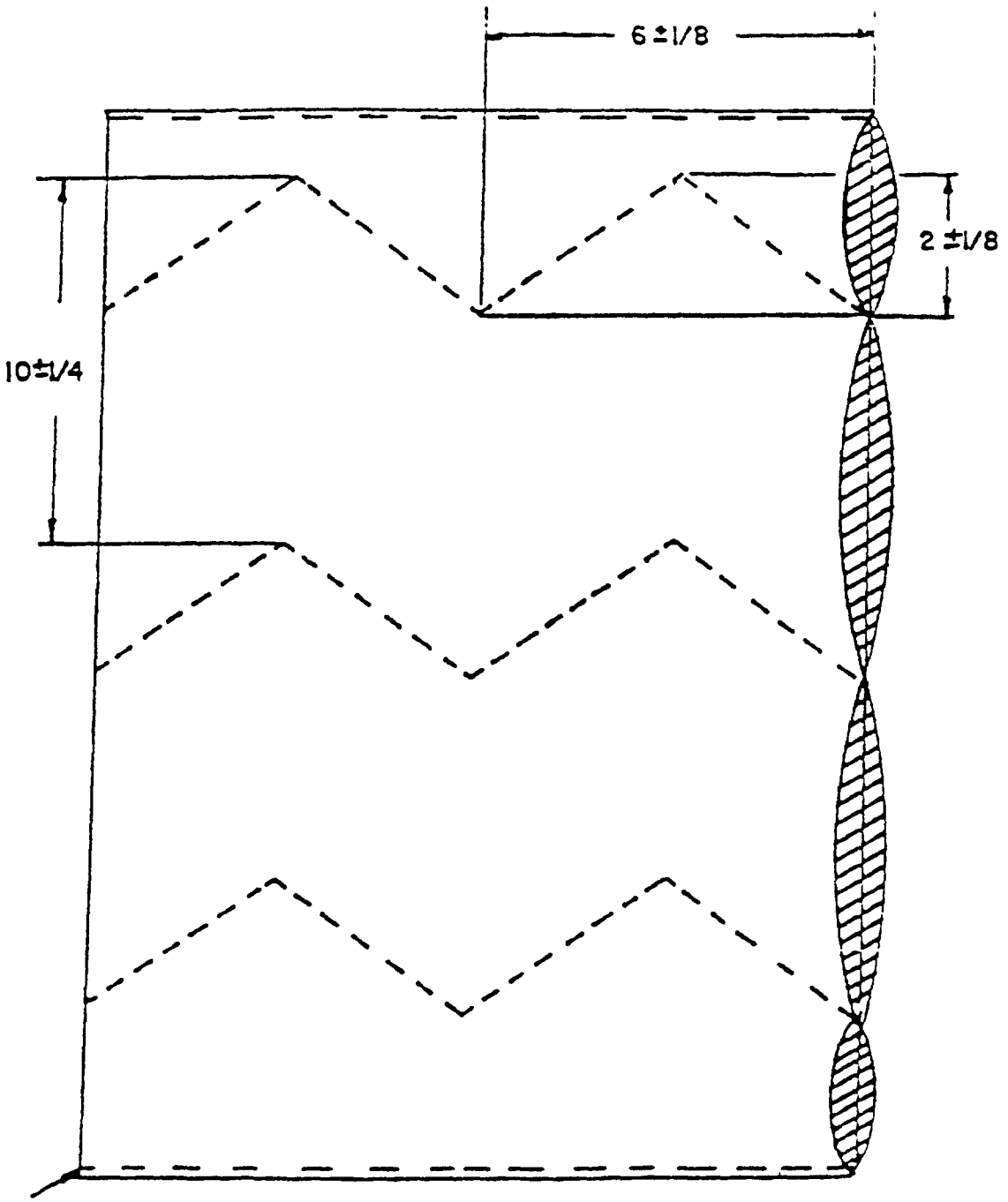


FIGURE 2. Zig zag pattern.

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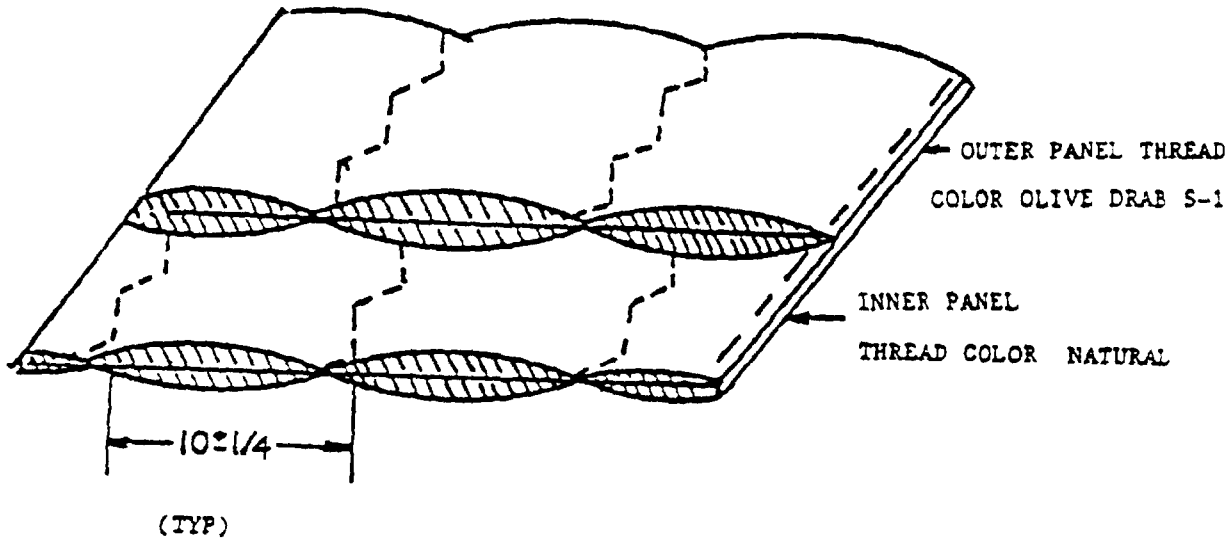


FIGURE 3. Quilt construction, zig zag pattern, offset quilt stitching rows.

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